

1) Znaleźć ekstremum lokalne funkcji 2 zmiennych

$$F(x, y) = \ln x + \ln(y^3) - xy - 4y^2 + \ln e^{10}$$

$$D: x > 0, y > 0$$

$$F'_x = \frac{1}{x} - y$$

$$F'_y = \frac{3y^2}{y^3} - x - 8y = \frac{3}{y} - x - 8y$$

$$F''_{xx} = -\frac{1}{x^2}$$

$$F''_{yy} = -\frac{3}{y^2} - 8$$

$$F''_{xy} = -1$$

$$F''_{yx} = -1$$

$$F'_x = 0 \quad \wedge \quad F'_y = 0$$

$$\begin{cases} \frac{1}{x} - y = 0 \\ \frac{3}{y} - x - 8y = 0 \end{cases} \Rightarrow y = \frac{1}{x}$$

$$3x - x - \frac{8}{x} = 0$$

$$2x - \frac{8}{x} = 0 \quad || \cdot x$$

$$2x^2 - 8 = 0$$

$$x^2 - 4 = 0$$

$$(x-2)(x+2) = 0$$

$$\begin{cases} x_1 = 2 \\ y_1 = \frac{1}{2} \end{cases} \quad \vee \quad x_2 = -2 \notin D$$

$$H_{(2, \frac{1}{2})} = \begin{vmatrix} \overset{0}{\left(\frac{1}{4} \right)} & -1 \\ -1 & -20 \end{vmatrix} =$$

$$= 5 - 1 = 4 > 0$$

Max lokalne w punkcie $(2, \frac{1}{2})$

$$\int \frac{dx}{1+\sin^2 x} = \int \frac{\cos x}{\sin x (1+\sin^2 x)} dx = \left\{ \begin{array}{l} t = \sin x \\ dt = \cos x dx \\ dx = \frac{dt}{\cos x} \end{array} \right\} =$$

$$= \int \frac{1}{t(1+t^2)} dt = \int \frac{1}{t} - \frac{t}{1+t^2} dt =$$

$$\frac{1}{t(1+t^2)} = \frac{A}{t} + \frac{Bt+C}{1+t^2} = \frac{A+Bt^2+Ct}{t(1+t^2)}$$

$$A+B=0 \Rightarrow B=-1$$

$$Ct=0 \Rightarrow C=0$$

$$A=1$$

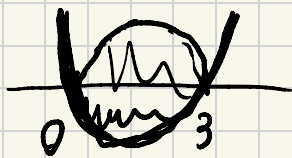
$$\frac{1}{t^2(1+t^2)} = \frac{A}{t} + \frac{B}{t^2} + \frac{Ct+D}{1+t^2}$$

$$\Rightarrow \int \frac{1}{t} dt - \int \frac{t}{1+t^2} dt =$$

$$= \left\{ \begin{array}{ll} f=t & f'=1 \\ g'=\frac{1}{1+t^2} & g=\arctan t \end{array} \right\} =$$

$$= \int \frac{1}{t} dt - (t \arctan t - \int \arctan t dt)$$

$$\int_{-4}^6 x |x^2 - 3x| dx =$$



$$= \int_{-4}^0 x(x^2 - 3x) dx + \int_0^3 x(x^2 - 3x) dx + \int_3^6 x(x^2 - 3x) dx =$$

$$= \int_{-4}^0 x^3 - 3x^2 dx - \int_0^3 x^3 - 3x^2 dx + \int_3^6 x^3 - 3x^2 dx =$$

$$= \left(\frac{1}{4}x^4 - 9x^3 \right) \Big|_{-4}^0 - \left(\frac{1}{4}x^4 - 9x^3 \right) \Big|_0^3 + \left(\frac{1}{4}x^4 - 9x^3 \right) \Big|_3^6$$